

Use-Case Flow Specification

Problem Statement #19: Personalized Meal & Diet Subscription Manager

Use Case ID / Name	UC-02: Generate Weekly Meal Plan
Primary Actor	Subscriber (system-triggered on a weekly schedule; reviewed by Dietitian)
Secondary Actors	Dietitian, Notification Service
Trigger	Weekly plan-generation schedule is reached, or the Subscriber updates their dietary profile before the cutoff.

Preconditions

- The Subscriber has an active subscription and a completed dietary profile (allergies, diabetic limits, calorie goals).
- The Subscriber's profile has passed initial validation (no missing mandatory fields).

Postconditions

- Success: A weekly meal plan exists in 'Released' status, approved by a Dietitian, and visible to the Subscriber.
- Failure: No plan is released; the Subscriber's existing (prior week's) plan or a fallback safe-menu remains active.

Main Success Scenario

1. Subscriber (or the weekly scheduler) initiates weekly meal plan generation.
2. System retrieves the Subscriber's current dietary profile (allergens, diabetic limits, calorie goals).
3. System includes UC-07 (Validate Allergen & Macro Constraints) to filter the recipe catalog against the profile.
4. System composes a 7-day menu from the filtered recipe set, balancing calorie and macro-nutrient targets per day.
5. System sets the plan status to 'Draft' and notifies the assigned Dietitian for review.
6. Dietitian reviews the draft plan and records an 'Approved' decision.
7. System sets the plan status to 'Released' and notifies the Subscriber that the new weekly menu is available.
8. Use case ends successfully.

Alternate Flow

6a. Dietitian Rejects the Draft Plan

- 6a1. Dietitian marks the draft plan 'Rejected' and enters a reason (e.g., insufficient calorie variance).
- 6a2. System returns to step 3 and regenerates the plan, excluding the previously rejected recipe combinations.
- 6a3. If two consecutive drafts are rejected, the system escalates to a senior Dietitian and keeps the Subscriber's prior week's plan active until resolved.